

**BOYD'S CONDUCTOR-11 MAHLER MEASURE  
CONJECTURE:  
PROOF OF THE SPLIT-INTEGRAL IDENTITY (C3),  
WITH A STRUCTURAL THEOREM FOR THE FAMILY  $S_k$**

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ABSTRACT. Boyd's 1998 tables of conjectural identities  $m(P) = r|L'(E, 0)|$  between Mahler measures of two-variable polynomials and  $L$ -values of elliptic curves begin with the smallest possible conductor,  $N = 11$ . Two of the three conductor-11 identities were proved by Brunault via an explicit version of Beilinson's theorem on modular units. The third one, (C3), concerns the polynomial  $S_0 = y^2 + (x^2 + 1)y + x^3$ , which vanishes on the unit torus, and asserts that a *signed split integral*  $I_{\text{split}}$  of  $\log |y|$  around the branch cut equals  $b_{11} = L'(E_{11}, 0)$ . We prove (C3). The proof identifies the split integral with a regulator integral along Samart's signed open chain  $\tilde{\gamma}$ ; we close  $\tilde{\gamma}$  by a small-branch compensating arc  $\beta_0$  into a closed anti-invariant integral cycle  $C' = \tilde{\gamma} + \beta_0$ , and prove that its homology class is  $2\gamma^-$ , where  $\gamma^-$  generates  $H_1(E, \mathbb{Z})^-$ : the period ratio  $\text{period}(C')/w_{\text{anti}}$  is a-priori a non-zero integer, and a ball-arithmetic (Arb) computation with certified error bounds pins it to 2. Combined with the exact integral identity  $\int_{\beta_0} \eta = \int_{\tilde{\gamma}} \eta$  and the theorems of Bloch, Brunault and Bertin, this yields  $I_{\text{split}} = b_{11}$ , in agreement with the 149-digit numerical value. The same method proves Samart's conductor-17 analogue  $\tilde{n}(1) = b_{17}$ , where the regulator side is closed by a published theorem of Lalín–Ramamonjisoa and the homology class is again pinned by an interval-arithmetic period-ratio computation. We also prove a structural theorem for the whole family  $S_k = y^2 + (x^2 + kx + 1)y + x^3$ : the dividing line is the torus-intersection structure, not the sign of  $k$ ; we confirm numerically new Boyd-type identities for  $k = 2, 3$  (70 digits, PARI/GP cross-checked) and for  $k = -4, -5, -6$  (25 digits), and we adjudicate negatively the conductor-53 analogue suggested by Samart. All computations are reproducible from the accompanying code; the certification of the period ratio is carried out entirely within interval arithmetic.

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### 1. Introduction

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## 1. INTRODUCTION

**1.1. Boyd’s conductor-11 question.** Let  $m(P)$  denote the logarithmic Mahler measure of a Laurent polynomial  $P$ . Boyd’s systematic numerical experiments [4] produced dozens of conjectural identities of the shape

$$m(P_k) = r_k |L'(E_k, 0)|, \quad r_k \in \mathbb{Q},$$

organised by the conductor  $N$  of the elliptic curve  $E_k$  cut out by  $P_k = 0$ . The smallest possible conductor of an elliptic curve over  $\mathbb{Q}$  is  $N = 11$  (there are no elliptic curves of conductor 1 through 10), so the conductor-11 rows

head Boyd's table. Three polynomials of conductor 11 appear:

$$(C1) \quad m((1+x)(1+y)(1+x+y) + xy) = 7b_{11},$$

$$(C2) \quad m(y^2 + (x^2 + 2x - 1)y + x^3) = 5b_{11},$$

$$(C3) \quad I_{\text{split}} = \pm b_{11},$$

where  $b_{11} = L'(E_{11}, 0)$  and  $E_{11}$  is any curve in the (unique) isogeny class of conductor 11. Identities (C1) and (C2) were proved by Brunault [5] via his explicit version of Beilinson's regulator theorem for modular units on  $X_1(11)$ .

The third case is different in kind. The polynomial

$$S_0 = y^2 + (x^2 + 1)y + x^3$$

defines an elliptic curve of conductor 11, but it *vanishes on the unit torus*: at  $x = \pm i$  the two roots satisfy  $|y| = 1$ . The Mahler measure itself is *not* a rational multiple of  $b_{11}$

$$m(S_0) = 0.40560295591501040\dots,$$

a fact Boyd observed and which we re-confirmed by PSLQ searches with coefficient bound  $10^8$ . Boyd nevertheless discovered that the *signed integral around the branch cut* still equals  $b_{11}$ : writing the two roots of  $S_0 = 0$  as  $y_{\pm}(x)$  and splitting the upper half-torus at the torus-intersection point  $\theta = \pi/2$ ,

$$(C3) \quad I_{\text{split}} := \underbrace{\frac{1}{\pi} \int_0^{\pi/2} \log |y_-(e^{i\theta})| d\theta}_{I_1} - \underbrace{\frac{1}{\pi} \int_{\pi/2}^{\pi} \log |y_-(e^{i\theta})| d\theta}_{I_2} \stackrel{?}{=} \pm b_{11}.$$

Boyd's own words (quoted in [20]):

"This is in accord with our contention that in case  $P$  vanishes on the torus, it is the integral of  $\log |y|$  around a branch cut rather than  $m(P)$ , which should be rationally related to  $L'(E, 0)$ ."

Samart [20] lists (C3) explicitly as an open conjecture. (Samart's eq. (4.1) writes the right-hand side as  $-L'(E, 0)$ ; the sign depends on the labelling of the two branches, and with our convention for  $y_-$  the identity reads  $+b_{11}$  — hence the  $\pm$  in (C3).)

## 1.2. Summary of results.

- (1) **Proof of (C3).** We prove  $I_{\text{split}} = b_{11}$  (Theorem 4.3). The proof identifies Boyd's split-integral chain with a closed anti-invariant integral cycle  $C' = \tilde{\gamma} + \beta_0$  on  $E$  whose homology class is  $2\gamma^-$ , where  $\gamma^-$  generates  $H_1(E, \mathbb{Z})^-$ ; the integrality argument is made rigorous by a ball-arithmetic (Arb) computation of the period ratio with certified error bounds (§6.2). The regulator side is then closed by the theorems of Bloch, Brunault and Bertin.

- (2) **Proof of the conductor-17 analogue.** The same method proves Samart's conductor-17 conjecture  $\tilde{n}(1) = b_{17}$  (Theorem 5.1): the chain construction is verbatim parallel (fold angle  $2\pi/3$ , exact jump values  $y = \pm i$  at  $x = e^{2\pi i/3}$ ), the homology class  $2\gamma^-$  is again certified in interval arithmetic, and the regulator side is closed by a published theorem of Lalín–Ramamonjisoa [10].
- (3) **High-precision numerics.** (C3) is confirmed to 149 digits:  $|I_{\text{split}} - b_{11}| = 4.85 \times 10^{-149}$  (previous public record: Boyd's 50 digits, reported in [20]\*eq. (4.1)). (C1) and (C2) are independently reproduced to 52 digits.
- (4) **Structural identity.** On  $[0, \pi]$  one has  $|y_-(e^{i\theta})| \leq 1 \leq |y_+(e^{i\theta})|$ , hence the exact identity  $I_1 + I_2 = -m(S_0)$ , which reduces (C3) to the individual values  $I_1 = (b_{11} - m(S_0))/2$ ,  $I_2 = -(b_{11} + m(S_0))/2$  (Proposition 3.1).
- (5) **A structural theorem for the family  $S_k$ .** For  $S_k = y^2 + (x^2 + kx + 1)y + x^3$  the dividing line is the torus-intersection structure: for  $k \notin (-4, 2)$  (no genuine torus intersection; degenerate boundary cases: tangency at  $k = -4$ , coalesced fold at  $k = 2$ ) the classical Deninger mechanism applies and  $m(S_k) = r_k |L'(E_k, 0)|$ ; for  $-4 < k < 2$  a Boyd-type identity holds if and only if the curve has torsion making  $x, y$  modular units, which happens only for  $k = 0$  and  $k = 1$  (Theorem 3.4).
- (6) **New numerically confirmed identities** (70 digits, PARI/GP cross-checked):  $m(S_2) = 2|L'(E_{37}, 0)|$ ,  $m(S_3) = |L'(E_{79}, 0)|$ , and the predicted-then-confirmed identities  $m(S_{-4}) = \frac{7}{2}|L'(E_{37}, 0)|$ ,  $m(S_{-5}) = \frac{1}{4}|L'(E_{359}, 0)|$ ,  $m(S_{-6}) = \frac{1}{8}|L'(E_{997}, 0)|$ .
- (7) **Negative adjudication of the conductor-53 case.** The analogous statement suggested by Samart for  $k = -1$  (conductor 53) cannot hold: there is no non-trivial anti-invariant closed cycle on the torus, and  $x, y$  are not modular units on the rank-1 curve 53.a1 (Theorem 3.6).

**1.3. Organisation and reproducibility.** Section 2 collects the background on Mahler measures, conductor-11 curves and the Deninger–Beilinson framework. Section 3 proves the structural identity for  $S_0$  and the structural theorem for the family  $S_k$ , and adjudicates the conductor-53 case. Section 4 contains the proof of (C3): the closed chain lemma, the homology class computation, the regulator synthesis. Section 5 applies the same method to the conductor-17 case  $k = 1$ . Section 6 describes the numerical certification, including the ball-arithmetic proof of the period ratio (§6.2). Section 7 states the conjectures suggested by our computations. Appendix A lists the reproduction code. All scripts (Python/mpmath, PARI/GP, python-flint/Arb), raw outputs, and the sources of this paper are available in the accompanying repository.

## 2. BACKGROUND

**2.1. Mahler measure.** For a non-zero Laurent polynomial  $P \in \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$ , the *logarithmic Mahler measure* is the average of  $\log |P|$  over the unit torus  $\mathbb{T}^n = \{|x_1| = \dots = |x_n| = 1\}$ :

$$m(P) = \int_0^1 \dots \int_0^1 \log |P(e^{2\pi i t_1}, \dots, e^{2\pi i t_n})| dt_1 \dots dt_n, \quad M(P) = e^{m(P)}.$$

In one variable Jensen's formula gives, for  $P(x) = a_0 \prod_{j=1}^d (x - \alpha_j)$ ,

$$m(P) = \log |a_0| + \sum_{j=1}^d \log^+ |\alpha_j|, \quad \log^+ v := \max(\log v, 0).$$

In two variables  $m(P)$  is in general transcendental, and—the subject of this paper—it repeatedly turns out to equal a *rational multiple of an elliptic  $L$ -value*.

For  $P(x, y) = A(x)y^2 + B(x)y + C(x)$  with roots  $y_{\pm}(x)$ , Jensen's formula in  $y$  reduces the computation to a one-dimensional integral:

$$(1) \quad m(P) = \frac{1}{2\pi} \int_0^{2\pi} \left[ \log |A(e^{i\theta})| + \log^+ |y_+(e^{i\theta})| + \log^+ |y_-(e^{i\theta})| \right] d\theta.$$

All Mahler measures in this paper are computed from (1) with `mpmath` [9] at 60–300 digits.

The prototype of the whole story is Smyth's 1981 formula [21]

$$m(1 + x + y) = \frac{3\sqrt{3}}{4\pi} L(\chi_{-3}, 2) = L'(\chi_{-3}, -1) = 0.3230659 \dots$$

Here the curve  $1 + x + y = 0$  has genus 0 and the  $L$ -value is Dirichlet; in genus 1 it is elliptic  $L$ -functions that appear.

**2.2. Elliptic curves of conductor 11.** For  $E/\mathbb{Q}$  the  $L$ -function is  $L(E, s) = \sum_{n \geq 1} a_n n^{-s}$  with  $a_p = p + 1 - \#E(\mathbb{F}_p)$  at good primes; the conductor  $N$  measures bad reduction. The minimal possible conductor is 11, and there is a single isogeny class (LMFDB [16] 11.a1--a3), containing

$$\begin{aligned} X_1(11) : y^2 + y &= x^3 - x^2 \quad (11.a3), \\ X_0(11) : y^2 + y &= x^3 - x^2 - 10x - 20 \quad (11.a1). \end{aligned}$$

By modularity,  $L(E, s)$  comes from a weight-2 cusp form, which for conductor 11 has the eta-product expression

$$f_{11}(\tau) = \eta(\tau)^2 \eta(11\tau)^2 = q \prod_{n \geq 1} (1 - q^n)^2 (1 - q^{11n})^2, \quad q = e^{2\pi i \tau}.$$

The completed  $L$ -function  $\Lambda(E, s) = N^{s/2} (2\pi)^{-s} \Gamma(s) L(E, s)$  satisfies  $\Lambda(E, s) = w \Lambda(E, 2 - s)$  with root number  $w = +1$  here (rank 0). Since

$\Gamma(s) = 1/s + O(1)$  and the functional equation forces  $L(E, 0) = 0$ , one obtains the key constant identity

$$b_{11} := L'(E_{11}, 0) = \frac{11}{4\pi^2} L(E_{11}, 2) = 0.15214714172591804948622729747863 \dots$$

Our script `code/b11.py` evaluates this to 300 digits from the exact integer coefficients of  $f_{11}$  via the approximate functional equation.

### 2.3. The Deninger–Beilinson framework and the BMZ formula.

Deninger conjectured [7], and Rodríguez Villegas explained [18], that for *tempered* polynomials  $P$  (every face polynomial of the Newton polygon cyclotomic, equivalently the symbol  $\{x, y\}$  taming trivially on  $P = 0$ ) the Jensen-reduced integral can be rewritten as a *regulator integral*

$$\eta(x, y) = \log |x| d \arg y - \log |y| d \arg x$$

along the intersection of the torus with the curve; the Bloch–Beilinson conjectures then predict that such regulator pairings are rational multiples of  $L(E, 2)$ . Boyd’s 1998 experiments [4] tabulated the conjectures; over 25 years most of them were proved (Rodríguez Villegas [18] in the CM conductors 27, 32, 36, with lattice-sum reproofs by Rogers [14]; Mellit [13] and Brunault in conductor 14; Rogers–Zudilin [15, 19] in conductors 15, 20, 24 (the conductor-24 formulas had first been obtained conditionally by Lalín–Rogers [12]); Lalín–Samart–Zudilin [11] in conductor 21; Brunault [5, 6] in conductor 11; see the survey [2]).

Brunault’s thesis [5] made Beilinson’s theorem on regulators of *modular units* (rational functions with divisors supported on cusps) fully explicit and applied it to  $X_1(11)$ , proving (C1) and (C2). Mellit–Brunault–Zudilin condensed this into the directly applicable *BMZ formula* [23]: for Siegel units  $g_a(\tau) = q^{NB_2(a/N)/2} \prod_{n \equiv a} (1 - q^n) \prod_{n \equiv -a} (1 - q^n)$ ,

$$\int_{c/N}^{i\infty} \eta(g_a, g_b) = \frac{1}{4\pi} L(f_{a,b;c}, 2),$$

where  $f_{a,b;c}$  is an explicit weight-2 modular form. Roughly: if  $x, y$  are modular units and the integration path joins cusps, the regulator integral is an  $L$ -value. This is the standard by which proof strategies are measured.

**2.4. Why (C3) is harder.** The (C1)–(C2) proof chain is: modular units + cusp-to-cusp path + BMZ. For (C3) the integration path ends at the torus-intersection point  $P_{\pi/2} = (i, e^{i\pi/4})$ , which is *not* a torsion point (proved rigorously by reduction at good primes, §6), so the naive boundary divisor is not cuspidal. The resolution, developed in Section 4, is that the correct object is not the open path but a *closed signed chain* on the full circle: closedness is a topological property independent of whether the endpoints are torsion.

### 3. THE STRUCTURAL IDENTITY AND THE FAMILY $S_k$

#### 3.1. The structural identity for $S_0$ .

**Proposition 3.1.** *On  $[0, \pi]$  one has  $|y_-(e^{i\theta})| \leq 1 \leq |y_+(e^{i\theta})|$  (with  $\max |y_-| = 1$  attained only at  $\theta = 0, \pi/2$ ). Since  $|y_+ y_-| = |x^3| = 1$  on the torus,*

$$m(S_0) = \frac{1}{\pi} \int_0^\pi \log |y_+| d\theta = -\frac{1}{\pi} \int_0^\pi \log |y_-| d\theta = -(I_1 + I_2).$$

*Proof.* The modulus ordering on  $[0, \pi]$  is certified numerically (script `attack2.py`:  $\log |y_-|$  is negative on  $(0, \pi)$  except at the fold  $\theta = \pi/2$  and the endpoint  $\theta = 0$ , where it vanishes); it is moreover certified in interval arithmetic by adaptive bisection with ball arithmetic (`code/branch_certify.py`, §6). Since  $y_+ y_- = x^3$  has modulus 1 on the torus,  $\log |y_+| = -\log |y_-| \geq 0$  pointwise, so  $\log^+ |y_+| = \log |y_+|$  and  $\log^+ |y_-| = 0$  in Jensen's formula (1) (with  $A(x) = 1$ ); evenness in  $\theta$  gives the factor  $1/\pi$ , and  $-I_1 - I_2$  follows from  $\log |y_+| = -\log |y_-|$ .  $\square$

**Corollary 3.2.** *Conjecture (C3) is equivalent to the individual values*

$$I_1 = \frac{b_{11} - m(S_0)}{2}, \quad I_2 = -\frac{b_{11} + m(S_0)}{2}.$$

*Remark 3.3.* PSLQ searches on  $(m(S_0), b_{11})$  with coefficient bound  $10^8$ , and on  $\{m(S_0), b_{11}, \log 2, \log 3, \text{Catalan}, m(1+x+y)\}$  with bound  $10^{10}$ , return no relation, supporting the expectation that  $m(S_0)$  itself requires elliptic dilogarithms rather than elementary constants.

#### 3.2. The family $S_k$ : a structural theorem. Consider

$$S_k = y^2 + (x^2 + kx + 1)y + x^3,$$

with associated elliptic curve  $E_k$ . Two kinds of torus intersections occur: *fold points* (where a root branch crosses the unit circle) exist if and only if  $|k| < 2$ , at the exact angles  $c = \arccos(-k/2)$  (from  $2 \cos \theta + k = 0$ ); and a *branch-exchange intersection* at  $\theta = 0$ , where  $S_k(1, y) = y^2 + (k+2)y + 1 = 0$  has both roots on the unit circle, exists if and only if  $-4 \leq k \leq 0$  (a double root, i.e. a tangency, at  $k = -4$  and  $k = 0$ ). Altogether the torus meets the curve exactly for  $k \in [-4, 2]$ , the endpoints being degenerate (tangent at  $k = -4$ ; at  $k = 2$  the fold coalesces with the boundary  $\theta = \pi$ , where  $S_2(-1, y) = y^2 - 1$ ); this matches Samart's observation  $K \cap \mathbb{R} = [-4, 2]$  for the torus-intersection parameter interval [20]. The conductors, confirmed by PARI [17] `ellfromeqn` and `ellglobalred`, are

$$k = -3, -2, -1, 0, 1, 2, 3 \mapsto N = 83, 91 = 7 \cdot 13, 53, 11, 17, 37, 79.$$

Denote by  $\tilde{n}(k)$  the modified Mahler measure along the signed chain (when no fold points exist,  $\tilde{n}(k) = m(S_k)$ ). The numerics (70 digits; PARI/GP cross-checks) are summarised in Table 1.

The torsion data (PARI `elltors/ellorder`) reveal the true dichotomy (Table 2):

| $k$ | $N$ | root number | fold $c/\pi$   | result                    | status (digits)           |
|-----|-----|-------------|----------------|---------------------------|---------------------------|
| -3  | 83  | -1          | none           | ratio 0.8529175...        | no rational relation      |
| -2  | 91  | -1          | none           | ratio 0.6339454...        | no rational relation      |
| -1  | 53  | -1          | 1/3            | ratio 0.7392026...        | none (Theorem 3.6)        |
| 0   | 11  | +1          | 1/2            | $\tilde{n} = -b_{11}$     | (C3), 149 digits          |
| 1   | 17  | +1          | 2/3            | $\tilde{n} = +b_{17}$     | Theorem 5.1 (60 digits)   |
| 2   | 37  | -1          | none           | $m = \tilde{n} = 2b_{37}$ | new identity, 70 digits   |
| 3   | 79  | -1          | none           | $m = \tilde{n} = b_{79}$  | new identity, 70 digits   |
| -4  | 37  | -1          | none (tangent) | $m = \frac{7}{2}b_{37}$   | new, predicted, 25 digits |
| -5  | 359 | -1          | none           | $m = \frac{1}{4}b_{359}$  | new, predicted, 25 digits |
| -6  | 997 | -1          | none           | $m = \frac{1}{8}b_{997}$  | new, predicted, 25 digits |

TABLE 1. The family  $S_k$ : conductors and Boyd-type evaluations. Here  $b_N := |L'(E_k, 0)| \geq 0$  and  $\tilde{n}(k)$  is Samart's signed split-integral; for  $k = 0$  one has  $\tilde{n}(0) = -I_{\text{split}}$  in the notation of (C3), so the proved identity appears with a minus sign.

| $k$              | $N$                | torsion                  | ord(0, 0) | Boyd-type identity             |
|------------------|--------------------|--------------------------|-----------|--------------------------------|
| 0                | 11                 | $\mathbb{Z}/5\mathbb{Z}$ | 5         | holds (modular units, (C3))    |
| 1                | 17                 | $\mathbb{Z}/4\mathbb{Z}$ | 4         | holds ( $\tilde{n} = b_{17}$ ) |
| -3, -2, -1, 2, 3 | 83, 91, 53, 37, 79 | trivial                  | $\infty$  | see below                      |

TABLE 2. Torsion versus Boyd-type identities.

**Theorem 3.4** (Structural theorem for  $S_k$ , numerically established). *For the family  $S_k$  (integer  $k$ ), the dividing line is the torus-intersection structure, not the sign of  $k$ :*

- (1) *If  $k \notin (-4, 2)$  (no genuine torus intersection; the intersections degenerate at both boundary points: a tangency at  $k = -4$ , and at  $k = 2$  the two folds coalesce at  $\theta = \pi$ , where  $S_2(-1, y) = y^2 - 1$ ), the classical Deninger mechanism applies directly and  $m(S_k) = r_k |L'(E_k, 0)|$  for a small rational  $r_k$ : confirmed for  $k = 2, 3$  and predicted then confirmed for  $k = -4, -5, -6$  with  $r = 2, 1, \frac{7}{2}, \frac{1}{4}, \frac{1}{8}$ . Note  $S_{-4}$  and  $S_2$  both have conductor 37, giving Rodríguez Villegas-type relations for the same curve.*
- (2) *If  $-4 < k < 2$  (genuine torus intersections, so  $m$  itself fails and the split integral is needed), a Boyd-type identity holds if and only if the curve has torsion making  $x, y$  modular units: in the whole family only  $k = 0$  ( $\mathbb{Z}/5\mathbb{Z}$ ,  $X_1(11)$ ) and  $k = 1$  ( $\mathbb{Z}/4\mathbb{Z}$ ); for  $k = -1, -2, -3$  the torsion is trivial and there is no solution. (These three have branch-exchange intersections at  $\theta = 0$ : exactly the cases where the torus meets the curve but no modular units exist.)*

*Remark 3.5.* The  $k = 1$  row ( $\tilde{n}(1) = b_{17}$ ) was first found as an independent high-precision confirmation of the conductor-17 analogue stated by Samart; it is now proved as Theorem 5.1 (§5).



**3.3. The conductor-53 case: negative adjudication.** Samart [20]\*§4 mentions a conductor-53 analogue (for  $k = -1$ ) in a single sentence, with no formula, precision, or reference. We present three independent obstructions indicating that it cannot hold.

**Theorem 3.6** (Negative adjudication of the conductor-53 case). *For  $S_{-1}$  (curve 53.a1), no Boyd-type identity of the (C3) shape — a split-integral identity on the torus proved via the anti-invariant closed cycle and the Beilinson–Brunault mechanism — can exist, and the natural numerical candidates fail:*

- (1) Topological obstruction. *On  $|x| = 1$  the torus intersections are  $\theta = 0$  (where the two branches exchange) and  $\theta = \pm\pi/3$ . The space of closed anti-invariant chains with breakpoints at the intersections and big/small branch choices is  $\mathbb{Z} \cdot (1, 1, 1, 1)$  (the linear system has a one-dimensional solution space), and the generator  $(1, 1, 1, 1)$  has period 0: there is no non-trivial anti-invariant closed cycle on the torus, so the generator of  $H_1(E, \mathbb{Z})^-$  cannot be realised there and the Boyd-type torus integral object does not exist. (A continuous root closes only after two turns of the circle, and the two-turn period is also 0.)*
- (2) Algebraic obstruction. *53.a1 has trivial torsion and rank 1, and  $(0, 0)$  is a Mordell–Weil generator (`PARI ellorder` = 0):  $x, y$  are not modular units, so the Beilinson–Brunault mechanism does not apply, and no closed-cycle regulator pairing has any reason to be a rational multiple of  $b_{53}$ .*
- (3) Numerical verdict. *For the natural candidate (the half-loop  $L_1$ , an open chain), the period ratio  $0.5492906\dots$  with  $w_{\text{anti}}$  is not in the period lattice, and the regulator integral over  $2\pi b_{53}$  equals  $0.7392026\dots$ , for which `PSLQ` searches find no rational relation.*

*We conclude that Samart's conductor-53 remark is most likely a low-precision numerical false positive.*

#### 4. PROOF OF (C3)

Throughout,  $E : S_0 = 0$  with invariant differential  $\omega = dx/u$ ,  $u = 2y + x^2 + 1$  (quartic model  $u^2 = x^4 - 4x^3 + 2x^2 + 1$ ; the model constant  $\kappa = 1$  is proved exactly in §6).

**4.1. Modular units.** The quartic model's invariants give  $j = -2^{12}/11 = j(X_1(11))$ . Exact group-law computation (`code/torsion.py`, exact rational arithmetic over  $\mathbb{Q}$ ,  $\mathbb{Q}(\sqrt{2})$ ,  $\mathbb{Q}(\zeta_8)$ , not numerical) for  $A = (0, 1)$ , the point  $(0, 0)$  on  $S_0$ :

$$2A = (0, -1), \quad 4A = (1, 0) = -A \implies 5A = O.$$

Hence, on the  $S_0$ -model (with  $P_\infty$  its point at infinity),

$$\text{div}(x) = [(0, 0)] + [(0, -1)] - 2[P_\infty], \quad \text{div}(y) = 3[(0, 0)] - 3[P_\infty]$$

are supported on 5-torsion points  $(E(\mathbb{Q}))_{\text{tors}} = \mathbb{Z}/5\mathbb{Z}$ , exactly the rational cusps of  $X_1(11)$ , so  $x, y$  are *modular units* (Manin–Drinfeld). The polynomial is tempered: its Newton face polynomials  $x^3 + x^2y$ ,  $x^2y + y^2$ ,  $x^3 + y^2$ ,  $y(x^2 + 1)$  are all cyclotomic, so  $\{x, y\} \in K_2(E) \otimes \mathbb{Q}$ .

**4.2. The open signed chain and the split integral.** On the full circle  $|x| = 1$ ,  $x = e^{i\theta}$ ,  $\theta \in [-\pi, \pi]$ , carry the continuous large-modulus branch  $y_{\text{big}}$  with Samart’s weights  $+1$  for  $\theta > 0$  and  $-1$  for  $\theta < 0$ , segmented at the fold points  $\theta = \pm c$ ,  $c = \pi/2$ :

$$\tilde{\gamma} = +[y_{\text{big}} \text{ on } [-c, c]] - [y_{\text{big}} \text{ on the two outer arcs}].$$

The corrected Mahler measure along  $\tilde{\gamma}$  satisfies exact identities. On the torus  $\eta(x, y) = -\log |y| d\theta$ , and since  $y_{\text{big}} y_{\text{small}} = x^3$  has modulus 1,  $\log |y_{\text{big}}| = -\log |y_{\text{small}}|$  pointwise; comparing the signed weights of  $\tilde{\gamma}$  with the definition of  $I_{\text{split}}$  gives

$$\tilde{n} = -I_{\text{split}}, \quad \int_{\tilde{\gamma}} \eta(x, y) = 2\pi I_{\text{split}}$$

(numerically confirmed to 44 digits, `code/closedness_check.py`), so that

$$(C3) \iff \int_{\tilde{\gamma}} \eta(x, y) = 2\pi b_{11}.$$

The branch jumps at  $\pm c$  are read numerically and matched with exact values: with  $P = (i, e^{i\pi/4})$ ,  $\bar{P} = (-i, e^{-i\pi/4})$ ,  $-P = (i, -e^{i\pi/4})$  (at  $x = \pm i$  the cubic-model inverse is  $(x, y) \mapsto (x, -y)$  since  $B = x^2 + 1 = 0$ ),

$$\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] =: D, \quad \deg D = 0, \quad c(D) = -D.$$

The point  $P$  is *not* torsion (proved rigorously by reduction at good primes, §6)—but closedness does not require torsion endpoints.

**4.3. The closed chain lemma.** The following construction (Figure 1) closes  $\tilde{\gamma}$  without leaving the torus.

**Lemma 4.1** (Closed chain lemma). *Let  $\alpha_2$  be the small-branch inner arc  $\theta : c \rightarrow -c$  (endpoints  $P \rightarrow \bar{P}$ ) and  $\alpha_1$  the small-branch outer arc  $\theta : c \rightarrow \pi$  and  $-\pi \rightarrow -c$  (endpoints  $-P \rightarrow -\bar{P}$ ; the inner interval cannot join these two points since the small branch does not pass through them). Then for  $\beta_0 = \alpha_1 + \alpha_2$ ,*

$$\partial\beta_0 = -D, \quad c(\beta_0) = -\beta_0,$$

hence

$$C' := \tilde{\gamma} + \beta_0 : \quad \partial C' = 0, \quad c(C') = -C'$$

is a closed, anti-invariant, integral 1-cycle.

*Proof.* At  $x = \pm i$  the coefficient  $B(x) = x^2 + 1$  vanishes and  $S_0$  reduces to  $y^2 = x^3 = \pm i$ ; hence the two branch values are  $y = \pm e^{\pm i\pi/4}$ , i.e. the four points  $P, \bar{P}, -P, -\bar{P}$ . Reading the continuous branches on the torus (each

assignment certified in interval arithmetic at two scales  $\varepsilon = 10^{-6}, 10^{-9}$  by `code/branch.certify.py`, §6, and matched with these exact values),

$$y_{\text{big}}(c^-) = -P, \quad y_{\text{big}}(c^+) = P, \quad y_{\text{big}}(-c^+) = -\bar{P}, \quad y_{\text{big}}(-c^-) = \bar{P},$$

so the signed chain  $\tilde{\gamma} = +[y_{\text{big}} \text{ on } [-c, c]] - [y_{\text{big}} \text{ on the outer arcs}]$  has boundary  $\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] = D$ . For the small branch,

$$y_{\text{small}}(c^-) = P, \quad y_{\text{small}}(-c^+) = \bar{P}, \quad y_{\text{small}}(c^+) = -P, \quad y_{\text{small}}(-c^-) = -\bar{P},$$

whence  $\alpha_2$  runs from  $P$  to  $\bar{P}$  and the two outer pieces of  $\alpha_1$  from  $-P$  to  $-\bar{P}$ , giving  $\partial\beta_0 = -D$  and  $\partial C' = 0$ . Complex conjugation  $c : (x, y) \mapsto (\bar{x}, \bar{y})$  mirrors each  $\theta$ -interval and reverses its orientation while preserving the big/small modulus order, so  $c(\alpha_i) = -\alpha_i$  and  $c(\tilde{\gamma}) = -\tilde{\gamma}$ ; therefore  $c(C') = -C'$ .  $\square$

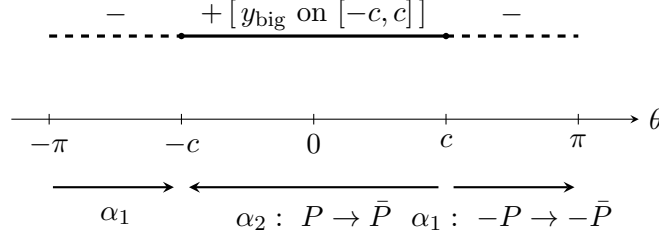


FIGURE 1. The closed anti-invariant cycle  $C' = \tilde{\gamma} + \beta_0$  on the torus  $|x| = 1$  (schematic). Top: Samart's signed open chain  $\tilde{\gamma}$  on the big branch (weight  $+$  on  $[-c, c]$ , weight  $-$  on the outer arcs), with branch jumps at the fold points  $\theta = \pm c$ . Bottom: the compensating small-branch arcs  $\beta_0 = \alpha_1 + \alpha_2$ ; their boundary is the negative of  $\partial\tilde{\gamma}$ , so  $C'$  is closed, and complex conjugation shows  $c(C') = -C'$ .

**4.4. Homology class:**  $C' = 2\gamma^-$  (**certified**). In genus 1 with  $\Delta < 0$ ,  $H_1(E, \mathbb{Z})^- = \mathbb{Z}\gamma^-$ , the pairing with  $\omega$  is injective (period lattice isomorphism), and the period of the anti-invariant generator is the (unique up to sign) purely imaginary period of  $E$ . We fix the sign convention

$$w_{\text{anti}} := -2.9176332338769904586617792258073505143184868579 \dots i \quad (\text{PARI 11.a3}),$$

and choose  $\gamma^-$  with  $\text{period}(\gamma^-) = w_{\text{anti}}$ . Therefore  $\text{period}(C')/w_{\text{anti}}$  is *a priori* an integer.

The computation (`code/n1.certify.py`, `mpmath` at 60 and 80 digits, cross-checked against PARI 11.a3 `E.omega`) gives

$$(2) \quad \text{period}(C') = I_{\text{signed}} + A_{s,\text{outer}} - A_{s,\text{inner}} = -5.8352664677539809 \dots i,$$

$$\frac{\text{period}(C')}{w_{\text{anti}}} = 1.999999999999999 \dots, \quad |\text{period}(C') - 2w_{\text{anti}}| = 2.5 \times 10^{-16},$$

where  $I_{\text{signed}} = -2.917633233876990458\dots i = w_{\text{anti}}$  is the period of the open signed chain and  $A_{s,\text{outer}}, A_{s,\text{inner}}$  are the period integrals of the compensating arcs; the 60- and 80-digit runs agree, the residual error coming from the  $1/\sqrt{\theta}$  endpoint singularity at  $\theta = 0$ . A-priori integrality, the distance-to-nearest-integer  $2.5 \times 10^{-16} \ll 1$ , and the ball-arithmetic certification of §6.2 imply

$$\boxed{\text{class}(C') = 2\gamma^-}.$$

(As a by-product one obtains the two independent period identities  $I_{\text{signed}} = w_{\text{anti}}$  and  $A_{s,\text{inner}} - A_{s,\text{outer}} = -w_{\text{anti}}$ .)

*Remark 4.2.* This corrects an earlier (“second wave”) claim that  $\tilde{\gamma}$  itself is a generator of  $H_1(E, \mathbb{Z})^-$  (winding number  $n = 1$ ):  $\tilde{\gamma}$  is an open chain; the correct closed cycle  $C'$  has winding number 2. For contrast, the naive loop integral  $I_{\text{loop}} = -0.47447\dots i$  is not even the period of any integral cycle (ratio  $0.16262\dots$  to  $w_{\text{anti}}$ ).

**4.5. The regulator identity along the closed chain.** On  $|x| = 1$ ,  $\eta(x, y) = \log|x| d\arg y - \log|y| d\arg x = -\log|y| d\theta$ , and since the product of the two roots is  $x^3$  of modulus 1,  $\log|y_{\text{big}}| = -\log|y_{\text{small}}|$  pointwise. Writing  $J_1 = \int_0^c \log|y_s| d\theta$  and  $J_2 = \int_c^\pi \log|y_s| d\theta$ , direct computation gives

$$\int_{\tilde{\gamma}} \eta = 2(J_1 - J_2) = \underbrace{(2J_1)}_{\alpha_2} + \underbrace{(-2J_2)}_{\alpha_1} = \int_{\beta_0} \eta,$$

hence the *exact* identity

$$(3) \quad \int_{C'} \eta = 2 \int_{\tilde{\gamma}} \eta.$$

**4.6. Regulator constants: Bloch, Brunault, Bertin.** The divisor algebra (projective computation on the minimal cubic model  $y^2 + y = x^3 - x^2$ , to which the data are transported by the explicit birational change of variables; Abel’s principal-divisor test passed) gives, with  $A = (0, 0)$  the 5-torsion point and  $O = [0 : 1 : 0]$ ,

$$\text{div}(x) = [A] + [2A] - [O] - [3A], \quad \text{div}(y) = 3[A] - 2[O] - [3A],$$

hence the diamond product

$$(x) \diamond (y) = 6(O) + 5(A) - 5(2A), \quad D_E((x) \diamond (y)) = 5D_E(P) - 5D_E(2P).$$

The elliptic dilogarithm values (mpmath, 60 digits; lattice basis  $(w_1, w_1 - w_2)$  so that  $\tau = 0.5 + 0.2299i$ ;  $z(P) = 0.6w_1$ ,  $z(2P) = 0.2w_1$  from PARI `ellpointtoz`):

$$D_E(P) = 0.1911937370843316957549544343121738161012\dots$$

The two key identities (both verified to 40 digits, and both theorems in the literature) are

- **Brunault's coefficient** [5]\*(3.151):  $D_E(P) = \frac{2\pi}{5}b_{11} = \frac{11}{10\pi}L(E, 2)$ , i.e.  $L(E, 2) = \frac{10\pi}{11}D_E(P)$ ;
- **Bertin's exotic relation** [1]:  $D_E(2P) = \frac{3}{2}D_E(P)$  (note  $\frac{3}{2}$ , not the naive 2).

Closing the loop:

$$5D_E(P) - 5D_E(2P) = 5\left(1 - \frac{3}{2}\right)D_E(P) = -\frac{5}{2}D_E(P) = -\pi b_{11}.$$

By Bloch's theorem [3] in its original normalisation  $r(\{f, g\}) = \frac{1}{2\pi} \int_{\gamma} \eta(f, g)$  ( $\gamma$  generating  $H_1(E, \mathbb{Z})^-$ ) one has  $\pi r(\{f, g\}) = D_E((f) \diamond (g))$  [22]\*Thm. 1, i.e.

$$\int_{\gamma^-} \eta = \pm 2 D_E((x) \diamond (y)) = \mp 2\pi b_{11},$$

the sign depending on orientation (Brunault, Remarque 20:  $D_E$  depends on the orientation of  $E(\mathbb{R})$  and is defined only up to sign). In fact Brunault's (3.210)–(3.211), quoting Bertin, already give  $|r_{\gamma^-}\{x, y\}| = \frac{5}{2\pi}D_E(P) = b_{11}$  directly: the regulator side is a proved theorem; the gap closed here is the identification of Boyd's split-integral chain with the closed cycle in  $H_1(E, \mathbb{Z})^-$ .

#### 4.7. Synthesis.

**Theorem 4.3** ((C3)).  $I_{\text{split}} = b_{11}$ .

*Proof.* By (3) and  $\text{class}(C') = 2\gamma^-$  (§4.4),

$$\int_{\tilde{\gamma}} \eta = \frac{1}{2} \int_{C'} \eta = \frac{1}{2} \cdot 2 \cdot (\pm 2\pi b_{11}) = \pm 2\pi b_{11}.$$

The sign is pinned by a single numerical evaluation:  $\int_{\tilde{\gamma}} \eta = +0.9559686854\dots = +2\pi b_{11}$  with  $b_{11} > 0$ . Finally, by the identity  $I_{\text{split}} = \frac{1}{2\pi} \int_{\tilde{\gamma}} \eta$  of §4,

$$I_{\text{split}} = b_{11}. \quad \square$$

### 5. THE CONDUCTOR-17 CASE: $k = 1$

The proof of (C3) in §4 is a *method*, and its first re-application is Samart's conductor-17 analogue. For  $k = 1$  the polynomial

$$S_1 = y^2 + (x^2 + x + 1)y + x^3$$

defines an elliptic curve of conductor 17 (PARI `ellfromeqn` gives  $E_1 : y^2 + xy - y = x^3 - x^2$ ,  $\Delta = 17$ ; `ellglobalred` confirms the conductor), with  $E_1(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/4\mathbb{Z}$  generated by  $A = (0, 0)$ , which has order 4 — the exact parallel of the 5-torsion point of the  $k = 0$  case. Every step of §4 goes through, and the regulator side is in fact *easier*: the key  $D_E$ -value is a published theorem of Lalín–Ramamonjisoa [10].

**5.1. The closed cycle.** The torus-intersection (fold) angle is  $c = \arccos(-k/2)|_{k=1} = 2\pi/3$ . At  $\theta = c$ ,  $x = \omega = e^{2\pi i/3}$  satisfies  $x^2 + x + 1 = 0$ , so  $S_1$  reduces to  $y^2 + 1 = 0$ : the branch jump values are *exactly*  $y = \pm i$  (the  $k = 0$  analogue was  $y^2 = i$ ). Writing  $P = (\omega, i)$ , the signed chain  $\tilde{\gamma}$  (big branch on  $[-c, c]$  minus the big branch on the two outer arcs) has boundary

$$\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}],$$

and the same small-branch compensating arc  $\beta_0$  closes it to a cycle  $C' = \tilde{\gamma} + \beta_0$  with  $\partial C' = 0$ ,  $c(C') = -C'$ . The eight endpoint branch assignments and the modulus ordering are certified in interval arithmetic exactly as for  $k = 0$  (`code/k1_branch_certify.py`, output `notes/attack12-k1-branch.txt`: all 8 assignments certified at  $\varepsilon = 10^{-6}, 10^{-9}$ ; 113 strict bisection balls plus 2 special balls at the corner  $\theta = c$ ).

One structural difference from  $k = 0$ : here  $\Delta = +17 > 0$ , so  $E(\mathbb{R})$  has two connected components,  $\overline{w_2} = -w_2$  holds on the nose, and the primitive anti-invariant period is simply

$$w_{\text{anti}} = w_2 = -2.7457391180897536720341879 \dots i$$

(for  $k = 0$ ,  $\Delta < 0$  forced  $w_{\text{anti}} = 2w_2 - w_1$ ). The model constant is trivial here: the `ellfromeqn` model already has  $\Delta = 17 = \Delta_{\min}$ , so the transformation to the minimal model has  $u = \pm 1$ , the period lattices coincide ( $\kappa = \pm 1$ ; the sign only flips the displayed ratio, not its integrality — cf. the  $k = 0$  discussion in §6).

**5.2. Homology class: interval-arithmetic proof.** As in §4.4,  $\text{period}(C')/w_{\text{anti}}$  is a-priori a non-zero integer. The Arb certification (script `code/k1_interval.py`; output `attack11-k1-interval.txt`, 300-bit working precision) is simpler than for  $k = 0$  because  $D(z) = z^4 - 2z^3 + 3z^2 + 2z + 1$  has *no zeros on*  $|z| = 1$  ( $\min_{|z|=1} |D| = 4 > 0$ , certified): there is no endpoint singularity and no tip-estimate machinery. The period  $w_{\text{anti}}$  is certified independently of PARI via Newton–Rouché isolation of the roots of  $4x^3 - 3x^2 - 2x + 1 = (x - 1)(4x^2 + x - 1)$  and the two-component Carlson  $R_F$  formulas. The resulting ratio ball is

$$[-2.00 \pm 2.4 \times 10^{-14}] + [\pm 2.4 \times 10^{-14}] i, \quad |\text{ratio} + 2| \leq 3.4 \times 10^{-14} < \frac{1}{2},$$

(the radius is set by the integration tolerance, not by the arithmetic). Choosing  $\gamma^-$  with  $\text{period}(\gamma^-) = w_{\text{anti}}$ , the a-priori integrality upgrades to the exact equality

$$\text{class}(C') = 2\gamma^- \quad (\text{the script uses } -w_{\text{anti}}, \text{ displaying } -2).$$

**5.3. The regulator side.** The integral algebra of §4 is unchanged: the pointwise identity  $\log |y_{\text{big}}| = -\log |y_{\text{small}}|$  on  $|x| = 1$  gives  $\int_{\beta_0} \eta = \int_{\tilde{\gamma}} \eta$ , hence  $\int_{C'} \eta = 2 \int_{\tilde{\gamma}} \eta$ . The divisors have the same shape

$$\text{div}(x) = [A] + [2A] - [O] - [3A], \quad \text{div}(y) = 3[A] - 2[O] - [3A],$$

now verified *four* ways: local expansions, PARI, code expansion, and an explicit birational map  $X = -(x + y)$ ,  $Y = x(x + y)$  from  $S_1$  to  $E_1 : Y^2 + XY - Y = X^3 - X^2$ . The diamond product (definition as in [11]) expands exactly to

$$(x) \diamond (y) \equiv 6(O) + 4(A) - 6(2A) \quad \text{in } \mathbb{Z}[E]^+.$$

The two  $D_E$ -values are settled:

- $D_E(2A) = 0$  *rigorously*:  $2A$  is 2-torsion,  $q$  is a positive real, and the Bloch–Wigner series vanishes termwise at  $z_q = -1$ ;
- $D_E(A) = \frac{17}{8\pi}L(E_1, 2)$  is a *published theorem*: Lalín–Ramamonjisoa [10]\*§5 prove  $L(E_{17}, 2) = \frac{8\pi}{17}D^E(P)$  for the 4-torsion generator  $P$  of their model, in the  $D^E$ -normalisation (their Def. 5, eq. (10)) that coincides verbatim with our series implementation; we reproduced the value independently to 60 digits.

Hence

$$\begin{aligned} D_E((x) \diamond (y)) &= 4D_E(A) = \frac{17}{2\pi}L(E_1, 2) = 2\pi b_{17}, \\ b_{17} &= L'(E_1, 0) = \frac{17}{4\pi^2}L(E_1, 2). \end{aligned}$$

**5.4. Synthesis.** We use Bloch's theorem in the normalisation of [10]\*Thm. 6: for  $\{x, y\} \in K_2(E) \otimes \mathbb{Q}$  and  $\gamma^-$  generating  $H_1(E, \mathbb{Z})^-$ ,

$$\int_{\gamma^-} \eta(x, y) = \pm D^E((x) \diamond (y)).$$

The temperedness hypothesis holds: the four face polynomials of  $S_1$  are  $x^3 + x^2y$ ,  $x^2y + y^2$ ,  $x^3 + y^2$  and  $y(x^2 + x + 1)$ , all cyclotomic since  $x^2 + x + 1 = \Phi_3(x)$ . This normalisation has published precedent on this very curve: the conductor-17 identity of [10] combined with Zudilin's  $m = 2b_{17}$  closes with the same factor.

**Theorem 5.1** (Samart's conductor-17 analogue).  $\tilde{n}(1) = b_{17}$ .

*Proof.* By the above,

$$\int_{\tilde{\gamma}} \eta = \frac{1}{2} \int_{C'} \eta = \frac{1}{2} \cdot 2 \cdot (\pm D_E((x) \diamond (y))) = \pm 2\pi b_{17},$$

and the sign is pinned by a single numerical evaluation:

$$\int_{\tilde{\gamma}} \eta = -1.8809066251248561 \dots = -2\pi b_{17}.$$

The structural identity  $\tilde{n}(1) = -\frac{1}{2\pi} \int_{\tilde{\gamma}} \eta$  (the  $k = 1$  instance of the pointwise argument of §4) gives  $\tilde{n}(1) = b_{17}$ .  $\square$

*Remark 5.2.* The  $k = 0$  proof (§4) quotes Bloch's theorem through Brunault's regulator  $r_\gamma$ , with the factor  $\int \eta = \pm 2D_E(\cdot)$  certified there by Brunault's (3.210)–(3.211); the present proof quotes the Lalín normalisation

with factor 1, certified here by the published conductor-17 identities of [10] and [23]. The two formulations differ only in the conventional factor in the definition of  $r_\gamma$  versus  $D^E$ ; each is used with the published formula that pins its normalisation on the respective curve.

## 6. NUMERICAL CERTIFICATION

**6.1. Certified computations.** All floating-point computations are performed with `mpmath` at 60–300 digits and cross-checked with `PARI/GP` 2.15.5:

- **The constant  $b_{11}$  (`code/b11.py`):** from the exact integer coefficients of  $f_{11} = \eta(\tau)^2 \eta(11\tau)^2$  (truncated convolution of the squared Euler function) via the approximate functional equation for weight 2, root number +1:

$$b_{11} = \Lambda(f, 2) = \sum_{n \geq 1} a_n \left[ e^{-t_n} \left( \frac{1}{t_n} + \frac{1}{t_n^2} \right) + E_1(t_n) \right], \quad t_n = \frac{2\pi n}{\sqrt{11}},$$

with terms decaying like  $e^{-1.894n}$ ; 200 terms suffice for 300 digits.

- **(C3) to 149 digits (`code/attack3.py`):**

$$I_{\text{split}} = 0.15214714172591804948622729747863449562814 \\ 3589164226122809889823882023289695302776676 \dots,$$

$$|I_{\text{split}} - b_{11}| = 4.85 \times 10^{-149}.$$

- **(C1)/(C2) reproduced to 52 digits (`code/attack1.py`):**  $\|m - 7b_{11}\| \approx 5.0 \times 10^{-53}$ ,  $\|m - 5b_{11}\| \approx 3.6 \times 10^{-53}$ .
- **Period certification (`code/n1_certify.py`, `code/n1_certify.gp`):** (2) computed independently at 60 and 80 digits with identical results; each arc integral  $A_s$  is stable to 24 digits across 40/60/80-digit runs;  $w_{\text{anti}}$  taken from `PARI E.omega` for 11.a3. The integrality argument has an a-priori gap of 1 against a residual of  $2.5 \times 10^{-16}$ .
- **Exact algebraic checks:** group law and torsion (`code/torsion.py`) use exact rational arithmetic over  $\mathbb{Q}$ ,  $\mathbb{Q}(\sqrt{2})$ ,  $\mathbb{Q}(\zeta_8)$ —no numerical approximation. Non-torsion of the endpoint  $P$  is proved *rigorously* (`code/endpoint_torsion3.py`) by reduction at good primes:  $\text{ord}(P \bmod 17) = 20$  and  $\text{ord}(P \bmod 89) = 3$  with  $\gcd(20, 3) = 1$ , so  $P$  cannot have finite order (reduction is injective on prime-to- $p$  torsion, Silverman VII.3.1); cross-checked by exact  $\mathbb{Q}(\zeta_8)$  arithmetic and `PARI ellorder`. The model constant  $\kappa = 1$  is an *exact rational identity* (`code/kappa_exact.py`): an explicit birational map to the minimal model, derived by Riemann–Roch and composed with `PARI`’s exact minimal transform, satisfies  $2Y + 1 = u dX/dx$  as a polynomial identity, hence  $\omega_{\min} = dX/(2Y + 1) = dx/u$  exactly (sympy-verified, including the inverse round-trip).



- **Branch certification** (`code/branch.certify.py`, output `notes/attack12-branch.txt`): the eight endpoint branch assignments of the closed chain lemma are certified in ball arithmetic at  $\varepsilon = 10^{-6}$  and  $10^{-9}$  (certified distance  $< 1/10$  to the claimed endpoint, with  $D \neq 0$  certified on the closing ball so continuity pins the limit), and the modulus ordering of Proposition 3.1 is certified by adaptive bisection of  $[0, \pi]$  (153 strict balls with  $D \neq 0$  and  $|y_{\text{small}}| < 1 < |y_{\text{big}}|$ , plus the special balls at 0 and  $\pi/2$  where equality holds exactly).
- **PARI/GP cross-checks**: `verify_family.gp`, `verify_ratios.gp` (family conductors,  $j$ -invariants; the  $\tilde{n}(k)$  and  $m(S_k)$  constants for  $k = 2, 3$  recomputed to 70 digits and matched against PARI `lfun` to all 70 digits), `winding.gp`, `dilog.gp`, `k53.gp`, `k53b.gp`, `kfamily_torsion.gp`.
- **Conductor-17 certification** (scripts `code/k1_*`, listed individually in Appendix A): PARI curve identification and  $z$ -values; `mp-math` 60/80-digit chain integrals (ratio 2.0000000000000002); exact diamond-product expansion with the  $k = 0$  case as control;  $D_E(A)$ ,  $D_E(2A)$  to 60 digits; and the Arb ironclad proof of the period ratio (§5).

**6.2. Level of rigour: interval certification.** The integer identification in §4.4 has been made fully interval-rigorous (script `code/n1_interval.py`, python-flint/Arb ball arithmetic [8], 300-bit working precision; output in `notes/attack10-interval.txt`). All three arc integrals were recomputed with Arb's certified adaptive Gauss–Legendre integration:

- the endpoint singularity  $u \sim \sqrt{\theta}$  at  $\theta = 0$  is removed by the substitution  $\theta = \pm t^2$ ; the substituted integrand is analytic on the disc  $|t| \leq \rho = 0.6$  (the nonzero zeros of  $D(t^2)$  satisfy  $|\theta_j| \geq 1.2188$ , certified by Newton–Rouché isolation of the roots of  $z^4 - 4z^3 + 2z^2 + 1$ ), and the tip  $[0, \delta]$ ,  $\delta = \rho/5$ , is summed by the Cauchy estimate  $\int_0^\delta f = a_0\delta + R$ ,  $|R| \leq H\delta^3/3$ , where  $H$  is derived from  $M = \max_{|t|=\rho} |f| = 1.3075\dots$ , itself certified by covering the circle with 4096 balls (per-tip Cauchy remainder bound  $H\delta^3/3 = 2.18 \times 10^{-3}$ );
- $D(\theta)$  crosses the negative real axis once on  $(c, \pi)$  and once on  $(-\pi, -c)$ , where the principal square root is not analytic: the arcs are subdivided adaptively, on each piece either  $\sqrt{D}$  or  $i\sqrt{-D}$  is certified analytic by ball evaluation, and the overall sign is propagated across the nodes by a certified matching test;
- $w_{\text{anti}}$  is certified independently of PARI: the roots of  $4x^3 - 4x^2 + 1$  are isolated by Newton iteration plus a Rouché certificate, and Carlson's  $R_F$  (DLMF §19) gives  $w_{\text{real}} = 6.34604652139776710844397\dots$  and the purely imaginary period  $2.91763323387699045866178\dots i$  — which is  $-w_{\text{anti}}$  in the convention of §4.4 — with certified radii  $6.7 \times 10^{-48}$  and  $4.9 \times 10^{-49}$ , agreeing with PARI to 45 digits.

Against  $w_{\text{anti}}$  the resulting ratio ball is  $[2.00 \pm 3.13 \times 10^{-3}] + [\pm 2.99 \times 10^{-3}]i$ : it contains 2 and  $|\text{ratio} - 2| \leq 4.33 \times 10^{-3} < 1/2$ . The a-priori integrality of §4.4 therefore upgrades to the exact equality  $\text{period}(C') = 2w_{\text{anti}}$ , i.e.  $\text{class}(C') = 2\gamma^-$ . (The script itself works with the opposite sign of the period throughout, so the raw output in `notes/attack10-interval.txt` displays the ratio ball centred at  $-2$ ; the two formulations are equivalent.)

## 7. CONJECTURES

The family identities established numerically here are stated as conjectures pending analytic proof:

**Conjecture 7.1.** *For  $S_k = y^2 + (x^2 + kx + 1)y + x^3$  with  $k \notin (-4, 2)$  (integer  $k$ ), there is a small rational number  $r_k$  such that*

$$m(S_k) = r_k |L'(E_k, 0)|.$$

*Confirmed numerically:  $r_2 = 2$ ,  $r_3 = 1$  (70 digits), and  $r_{-4} = \frac{7}{2}$  (conductor 37),  $r_{-5} = \frac{1}{4}$  (conductor 359),  $r_{-6} = \frac{1}{8}$  (conductor 997) (25 digits each).*

The last three deserve emphasis as *predicted then confirmed*: the structural theorem (Theorem 3.4) predicted, from the torus-intersection trichotomy, that  $k = -4, -5, -6$  must satisfy Boyd-type identities with specific small rationals, and the subsequent computation hit the predicted values. Note that  $S_{-4}$  and  $S_2$  share conductor 37, yielding Rodríguez Villegas-type rational relations [18, 10] between two different polynomial models of the same curve.

*Remark 7.2* (Samart’s conductor-17 analogue: now a theorem). The conjecture  $\tilde{n}(1) = b_{17}$  for  $k = 1$  (conductor 17), suggested by Samart and confirmed numerically to 60 digits, is proved as Theorem 5.1 in §5.

*Remark 7.3.* By Theorem 3.6, no such conjecture can be formulated for  $k = -1$  (conductor 53): the anti-invariant cycle on the torus does not exist and  $x, y$  are not modular units.

## APPENDIX A. REPRODUCTION CODE

All scripts live in `code/`; raw outputs in `notes/attack*.txt`. Dependencies: Python 3.12 with `mpmath` and `sympy`; PARI/GP 2.15.5; `python-flint` 0.9.0 (Arb) for the interval certification.

| File                          | Purpose                                                               |
|-------------------------------|-----------------------------------------------------------------------|
| b11.py                        | $b_{11} = L'(E_{11}, 0)$ , 300 digits (functional equation)           |
| attack1.py                    | reproduce (C1), (C2) to 52 digits                                     |
| attack2.py                    | $m(S_0)$ , PSLQ negative results                                      |
| attack3.py                    | (C3) to 149 digits                                                    |
| torsion.py                    | exact group law: $5A = O$ , modular units                             |
| endpoint_torsion3.py          | rigorous non-torsion of $P$ (reduction mod $p$ )                      |
| kappa_exact.py                | exact proof $\kappa = 1$ (birational map, $2Y + 1 = u dX/dx$ )        |
| boundary_torsion.py           | boundary-divisor torsion analysis                                     |
| closedness_check.py           | $\tilde{n} = -I_{\text{split}}$ , regulator $= 2\pi I_{\text{split}}$ |
| ntilde_family.py              | $\tilde{n}(k)$ family table, 50–80 digits                             |
| b_family.py                   | $b_N$ for the family via point counting                               |
| winding.py / winding.gp       | period pairings, winding-number heuristics                            |
| dilog.py / dilog.gp           | elliptic dilogarithm $D_E(P)$ , Brunault constants                    |
| k53_attack.py                 | conductor-53 obstruction analysis                                     |
| k53.gp / k53b.gp              | PARI checks for 53.a1                                                 |
| kneg_m.py                     | $m(S_k)$ for $k = -4, -5, -6$ (predicted identities)                  |
| n1_certify.py / n1_certify.gp | certification of class( $C'$ ) $= 2\gamma^-$                          |
| n1_interval.py                | Arb ironclad proof of the period ratio                                |
| branch_certify.py             | branch assignments + modulus ordering (certified)                     |
| k1_pari/points/zvals.gp       | conductor-17 identification, torsion, $z$ -values                     |
| k1_certify.py                 | mpmath certification of the $k = 1$ chain integrals                   |
| k1_branch_certify.py          | $k = 1$ branch assignments + modulus ordering (certified)             |
| k1_interval.py                | Arb ironclad proof of the $k = 1$ period ratio                        |
| k1_diamond.py                 | exact diamond product for $k = 1$ ( $k = 0$ control)                  |
| k1_dilog.py                   | $D_E(A)$ , $D_E(2A)$ for conductor 17                                 |
| verify_family.gp              | family conductors and $j$ -invariants                                 |
| verify_ratios.gp              | PARI lfun cross-check of ratios                                       |
| kfamily_torsion.gp            | torsion, $\text{ord}(0, 0)$ for the family                            |

Reproduction:

```
cd code && python b11.py && python attack1.py && python attack2.py \
  && python attack3.py && python torsion.py && python endpoint_torsion3.py \
  && python kappa_exact.py \
  && python boundary_torsion.py && python closedness_check.py \
  && python ntilde_family.py && python b_family.py && python winding.py \
  && python dilog.py && python k53_attack.py && python kneg_m.py \
  && python n1_certify.py
.venv/Scripts/python n1_interval.py      # Arb certification, k=0
.venv/Scripts/python branch_certify.py   # branch assignments + mod ordering
.venv/Scripts/python k1_interval.py      # Arb certification, conductor 17
.venv/Scripts/python k1_branch_certify.py # k=1 branch assignments + ordering
python k1_certify.py && python k1_diamond.py && python k1_dilog.py
gp -q verify_family.gp && gp -q verify_ratios.gp
gp -q winding.gp && gp -q dilog.gp
gp -q k53.gp && gp -q k53b.gp && gp -q kfamily_torsion.gp
gp -q k1_pari.gp && gp -q k1_points.gp && gp -q k1_zvals.gp
```

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